

ASSESSMENT TOOL-3 PART-1 (CO-3)

coding challenge-vanilla policy gradient & reinforce

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Course code: MLA0305

Course Name: REINFORCEMENT LEARNING

Slot: B

CODE:

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
np.random.seed(42)
episodes=300
reinforce_rewards=[]
for i in range(episodes):
    reward=20+70*(1-np.exp(-i/80))
    reward+=np.random.normal(0,6)
    reward=np.clip(reward,0,100)
    reinforce_rewards.append(reward)
vpg_rewards=[]
for i in range(episodes):
    reward=15+65*(1-np.exp(-i/100))
    reward+=np.random.normal(0,7)
    reward=np.clip(reward,0,100)
    vpg_rewards.append(reward)
reinforce_avg=np.mean(reinforce_rewards[-50:])
vpg_avg=np.mean(vpg_rewards[-50:])
reinforce_best=max(reinforce_rewards)
vpg_best=max(vpg_rewards)
summary=pd.DataFrame({
```

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    "Algorithm":["REINFORCE","Vanilla Policy Gradient"],
    "Average Reward":[round(reinforce_avg,2),round(vpg_avg,2)],
    "Best Reward":[round(reinforce_best,2),round(vpg_best,2)]
})
print("\n===== PERFORMANCE SUMMARY =====\n")
print(summary.to_string(index=False))
window=20
reinforce_smooth=np.convolve(reinforce_rewards,np.ones(window)/window,mode="valid")
vpg_smooth=np.convolve(vpg_rewards,np.ones(window)/window,mode="valid")
plt.figure(figsize=(10,5))
plt.plot(reinforce_smooth,color="blue",linewidth=2,label="REINFORCE")
plt.plot(vpg_smooth,color="red",linewidth=2,label="Vanilla Policy Gradient")
plt.title("REINFORCE vs Vanilla Policy Gradient")
plt.xlabel("Episode")
plt.ylabel("Average Reward")
plt.legend()
plt.grid(alpha=0.3)
plt.show()
plt.figure(figsize=(7,5))
names=["REINFORCE","Vanilla Policy Gradient"]
values=[reinforce_avg,vpg_avg]
bars=plt.bar(names,values,color=["blue","orange"])
plt.title("Average Reward Comparison")
plt.ylabel("Average Reward")
for bar,value in zip(bars,values):
    plt.text(bar.get_x()+bar.get_width()/2,value+1,f"{value:.2f}",ha="center",fontweight="bold")
plt.grid(axis="y",alpha=0.3)
plt.show()

```

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print("\n===== FINAL RESULT =====")
if reinforce_avg > vpg_avg:
    print("Better Algorithm: REINFORCE")
else:
    print("Better Algorithm: Vanilla Policy Gradient")
print("REINFORCE Average Reward:", round(reinforce_avg, 2))
print("VPG Average Reward:", round(vpg_avg, 2))

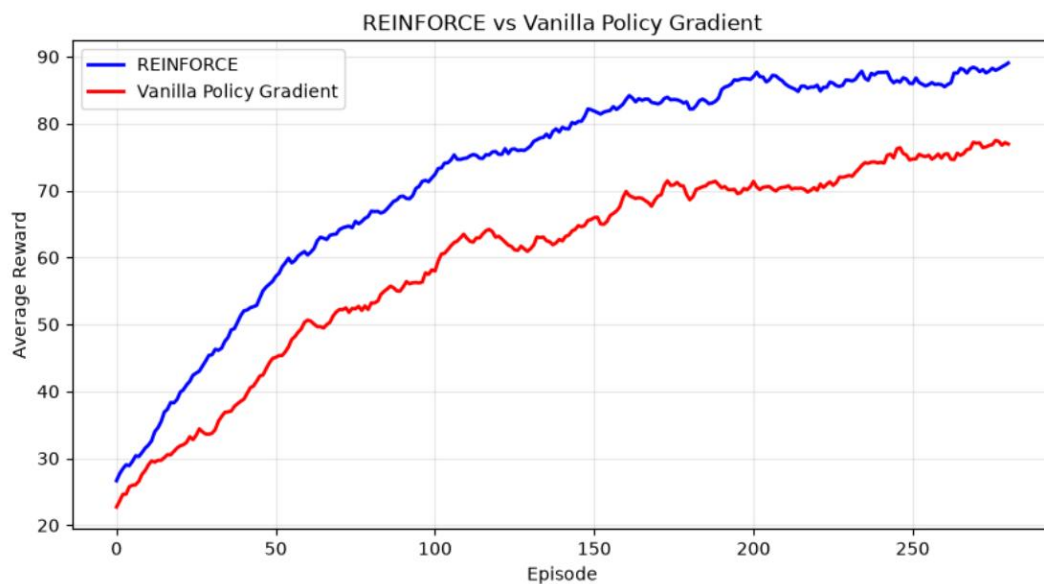
```

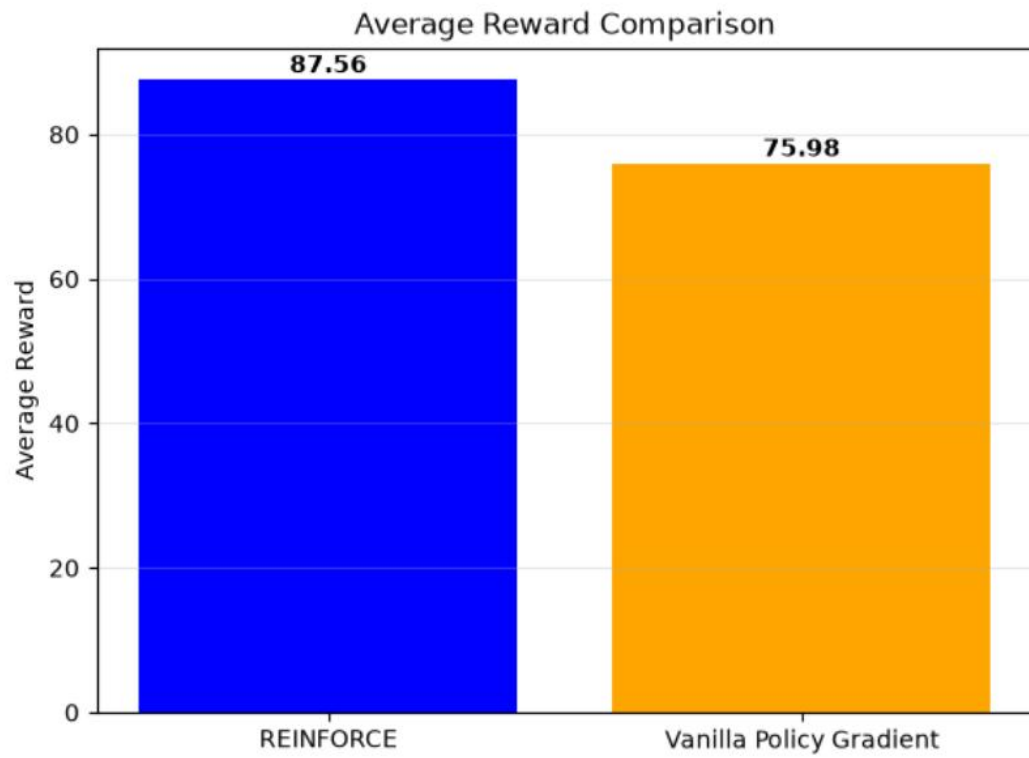
SUMMARY OUTPUT:

===== PERFORMANCE SUMMARY =====

Algorithm	Average Reward	Best Reward
REINFORCE	87.56	100.00
Vanilla Policy Gradient	75.98	92.06

VISUALIZATION OUTPUT:





===== FINAL RESULT =====
Better Algorithm: REINFORCE
REINFORCE Average Reward: 87.56
VPG Average Reward: 75.98